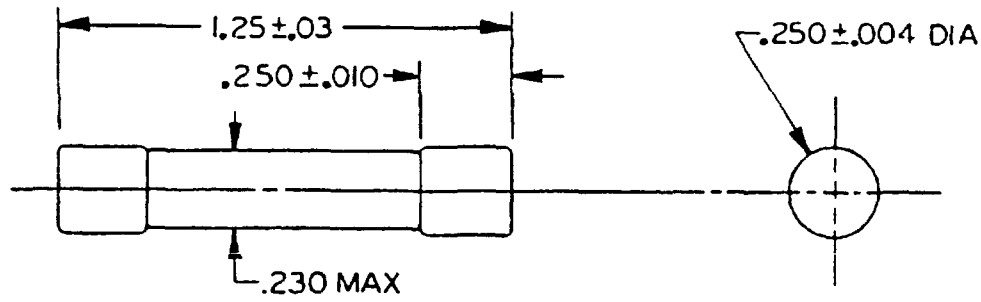
MILITARY SPECIFICATION SHEET

FUSE, CARTRIDGE, INSTRUMENT TYPE,

STYLE FM09, (NONINDICATING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the fuses described herein shall consist of this specification and the latest issue of MIL-F-23419.



Inches	mm
.004	0.10
.010	0.25
03	0.8
.230	5.84
.250	6 35
1.25	31 8

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Style FM09 fuses

Ⓐ denotes changes

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Applicable fuseholders

MIL-F-19207/1, Style FHL10U
MIL-F-19207/2, Style FHL11U
MIL-F-19207/8, Style FHL17G
MIL-F-19207/9, Style FHL18G
MIL-F-19207/11, Style FHN20G
MIL-F-19207/16, Style FHN26G
MIL-F-19207/17, Style FHN28WB
MIL-F-19207/18, Style FHL29G
MIL-F-19207/19, Style FHL30G
MIL-F-19207/21, Style FHL32W
MIL-F-19207/22, Style FHL33W
MIL-F-21346/3, Style FH22AM - shall be used for all testing.

Terminals: Ferrule type.
Material - Brass, copper, or copper alloy.
Finish - Nickel, bright alloy.
Strength - Test condition E; 2-pound-inch torque between ferrules and fuse body.

Body: Plastic or ceramic tube.

Current rating. See table I.

Characteristic. See table I.

Voltage rating: See table I.

(A) Current-carrying capacity for 20 amperes or less:

110 percent at room ambient temperature.

Characteristic A - 112.5 percent at -55°C, 91.5 percent at +85°C.

Characteristic B - 119 percent at -55°C, 69.5 percent at +85°C.

(A) Current-carrying capacity for 25 or 30 amperes:

100 percent with a maximum temperature rise of 90°C for all temperatures between -55°C and 85°C.

Characteristic: See table I.

Overload interrupt See table I (at all temperatures between -55°C and +85°C).

Resistance: See table I.

Short circuit interrupt: See table I.

Mechanical shock: Method I.

Thermal shock. Method 107, test condition A, MIL-STD-202.

Vibration. Characteristic A - all ratings, MIL-STD-202, method 204.

Characteristic B - Fuses rated 0 through 8/10 amperes shall meet the requirements of MIL-STD-202, method 201,

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TABLE I. Type designations and electrical ratings.

Type designation				Overload interrupt time 1/		AC short circuit interrupt		Cold resistance 2/ (ohms)	
Style	Characteristic	Maximum voltage	Current rating (amperes)	135% of rating-Maximum 1 hour	200% of rating (seconds)	300% of rating (seconds)	125 V	250 V	Minimum	Maximum
FM09	A	250 V	1/100 A		0.0001-1.0	0 0001-0.003	10,000 A	35 A	100.0	280.0
			1/32 A		0.0008-1.0	0.0002-0.003			20.0	50.0
			1/16 A		0.009-2.0	0.001-0.050			2 7	36.0
			1/8 A		0.01-2.0	0.003-0.15			.96	11.0
			1/4 A		0.02-2.0	0.01-0.20			1.98	4.62
			3/8 A		0.02-3.0	0.01-0.30			.96	2.24
			1/2 A		0 03-3.0	0.01-0.30			.66	1.54
			3/4 A		0.03-3.0	0 015-0.50			.20	.95
			1 A		0.05-3.0	0.015-0.50			.114	266
			1-1/4 A		0.05-3.0	0.02-0.50			.081	.200
			1-1/2 A		0.05-3.0	0.02-0.50			.061	.143
			2 A		0.08-3.0	0.04-0.50			.045	.106
			3 A		0.08-3.0	0.04-0.70			.023	.060
			4 A		0.10-5.0	0.05-1.00			.023	.055
			5 A		.20-60.0	0.05-6.00		100 A	.018	.042
			6 A		.20-60.0	0.05-6.00		100 A	.014	.032
			8 A		.30-60.0	0.05-6.00		100 A	.009	.021
			10 A		.30-60.0	0.05-6.00		100 A	.0068	.0158
			12 A		.30-60.0	0.05-6.00		200 A	.0045	.0114
			15 A		0.5-60.0	0.05-6.00		200 A	.0027	.0065
		125 V	20 A		0.5-60.0	0.05-6.00		Not rated	.0016	.0045
		125 V	25 A		0.5-60.0	0.05-6.00		Not rated	.0013	.0035
		125 V	30 A		0.5-60.0	0.05-6.00		Not rated	.0010	.0027
	B	250 V	1/100 A		4.0-250.0	2.0-40.0	10,000 A	35 A	950.0	4620.0
			1/32 A		4.0-80.0	2.0-25.0			100.0	462.0
			1/16 A		4.0-80.0	2.0-25.0			49.0	120.0
			1/10 A		4.0-80.0	2.0-25.0			19.9	46.6
			1/8 A		4.0-80.0	2.0-25.0			12.0	28.0
			15/100 A		4.0-80.0	2.0-25.0			9.1	21.4
			3/16 A		4.0-80.0	2 0-25.0			5.52	13.0
			1/4 A		4.0-80.0	2.0-25.0			3.18	8.0
			3/10 A		4.0-80.0	2.0-25.0			2.22	6.0
			3/8 A		4.0-80.0	2.0-25.0			1.5	4.77
			1/2 A		4.0-80.0	2.0-25.0			1.2	2.83
			6/10 A		4.0-80.0	2.0-25.0			.60	2.10
			3/4 A		4.0-80.0	2.0-25.0			.45	1.43
			8/10 A		4.0-80.0	2.0-25.0			.35	1.26
			1 A		4.0-80.0	2.0-25.0			.21	.49
			1-1/4 A		4.0-80.0	2.0-25.0			.16	.38
			1-1/2 A		4.0-80.0	2.0-25.0			.102	.340
			1-6/10 A		4.0-80.0	2.0-25.0			.099	.270
			2 A		4.0-80.0	2.0-25.0			.072	.170
			2-1/2 A		10.0-250.0	2.0-25.0			.066	.154
			2-8/10 A		10.0-250.0	2.0-25.0			.032	.090
			3 A		10.0-250.0	2.0-25.0			.026	.090
			3-2/10 A		10.0-250.0	2.0-25.0			.025	.090
		125 V	4 A		10.0-250.0	2.0-25.0		Not rated	.0240	.0560
			5 A		10.0-250.0	0.05-10.0			.0108	.0600
			6-1/4 A		10.0-250.0	0.05-10.0			.0071	.0450
			7 A		10.0-250.0	0 05-10.0			.0059	.0300
			8 A		10.0-250.0	0.05-10.0			.0070	0200
			10 A		10.0-250.0	0.05-10.0			.0056	.0140
			12 A		10.0-250.0	0.05-10.0			.0036	.0110
			15 A		10.0-250.0	0.05-10.0			.0017	.0080
			20 A		10.0-250.0	0.05-10.0			.0012	.0040
			25 A		10.0-250.0	0.05-10.0			.0009	.0030
			30 A		10.0-250.0	0.05-10.0			0006	0022

1/ The overload interrupt time ranges, as shown, cover the temperature range from -55°C to +85°C For characteristic B, the overload interrupt time at room ambient temperature shall be 12 seconds minimum at 200 percent of rating.

2/ Cold resistance is measured at 10 percent or less of rated current.

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Qualification inspection

Inspection routine. Group II samples shall be as shown in table II below:

TABLE II. Group II samples.

Inspection	Number of sample fuses
Group II	20
Terminal strength	20
Overload interrupt	
135% at room ambient	4
200% at -55°C	4
200% at +85°C	4
300% at -55°C	4
300% at +85°C	4
Insulation resistance	20

1/ For characteristic B fuses only, four additional fuses shall be tested at 200 percent of rating at room ambient temperature.

The number of group III samples for qualification (or first article) testing shall consist of four sample fuses of each of the types as shown in table III. Short circuit interrupt tests shall be conducted at the alternating current voltage indicated, and the fuses shall be capable of withstanding a minimum short circuit current as shown in table III. Insulation resistance test shall be as specified in MIL-F-23419.

TABLE III. Short circuit tests.

Fuse type	Current rating	Short circuit current	AC type voltage
FM09A	4 A	35 A	250
FM09A	10 A	100 A	250
FM09A	15 A	200 A	250
FM09A	30 A	10,000 A	125
FM09B	3 2/10 A	35 A	250
FM09B	30 A	10,000 A	125

Fuse selection. The following steps should apply in the selection of a fuse for any application

a. Step 1 - Select a fuse with a voltage rating equal to or in excess of the circuit voltage.

b Step 2 - The short circuit interrupting rating of the fuse should be checked to determine that it is adequate to protect against the short circuit currents that the particular equipment or circuit can be exposed to.

c. Step 3 - The ambient temperature in which the fuse is to be used should be compared to the curves shown on figure 1 or 2 to determine the effect the ambient will have on the current rating. For example, if a characteristic A fuse is to be used in a 60°C ambient, it will only carry 95 percent of its nominal rating A fuse which is used in a -20°C ambient will carry 107 percent of its nominal rating.

d. Step 4 - A fuse rating should be selected on the basis of equipment ampere load with compensation made for ambient temperature (see figures 2 and 3) Characteristics A and B fuses should not be loaded to more than 80 percent of the fuse rating (or 80 percent of the adjusted rating based on ambient temperature consideration) to avoid nuisance openings caused by inrush currents and other harmless overloads. Time lag (characteristic B) fuses can be loaded to a higher level because of the ability to override harmless overloads, transients, and inrush currents of short duration.

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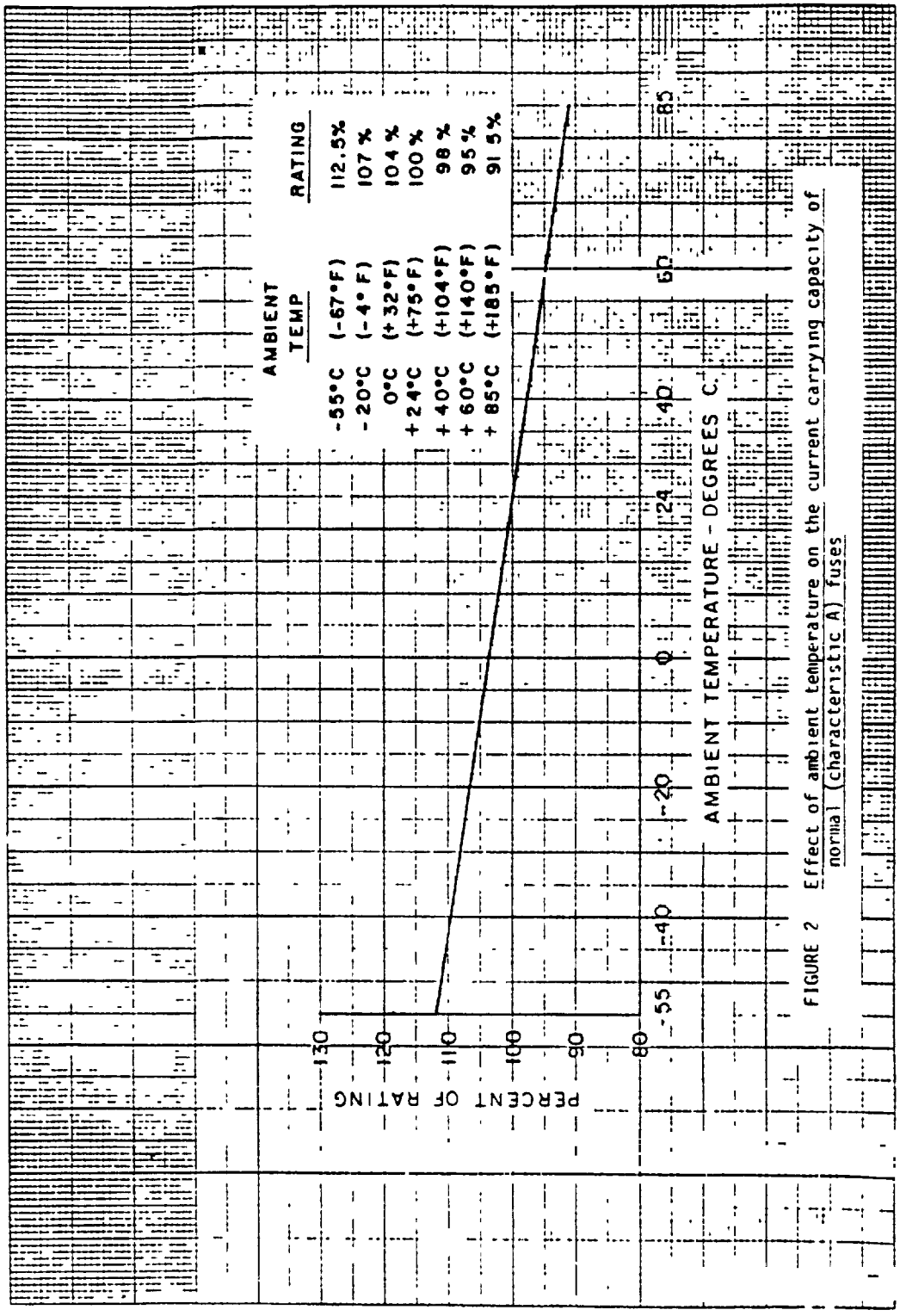


FIGURE 2 Effect of ambient temperature on the current carrying capacity of normal (characteristic A) fuses

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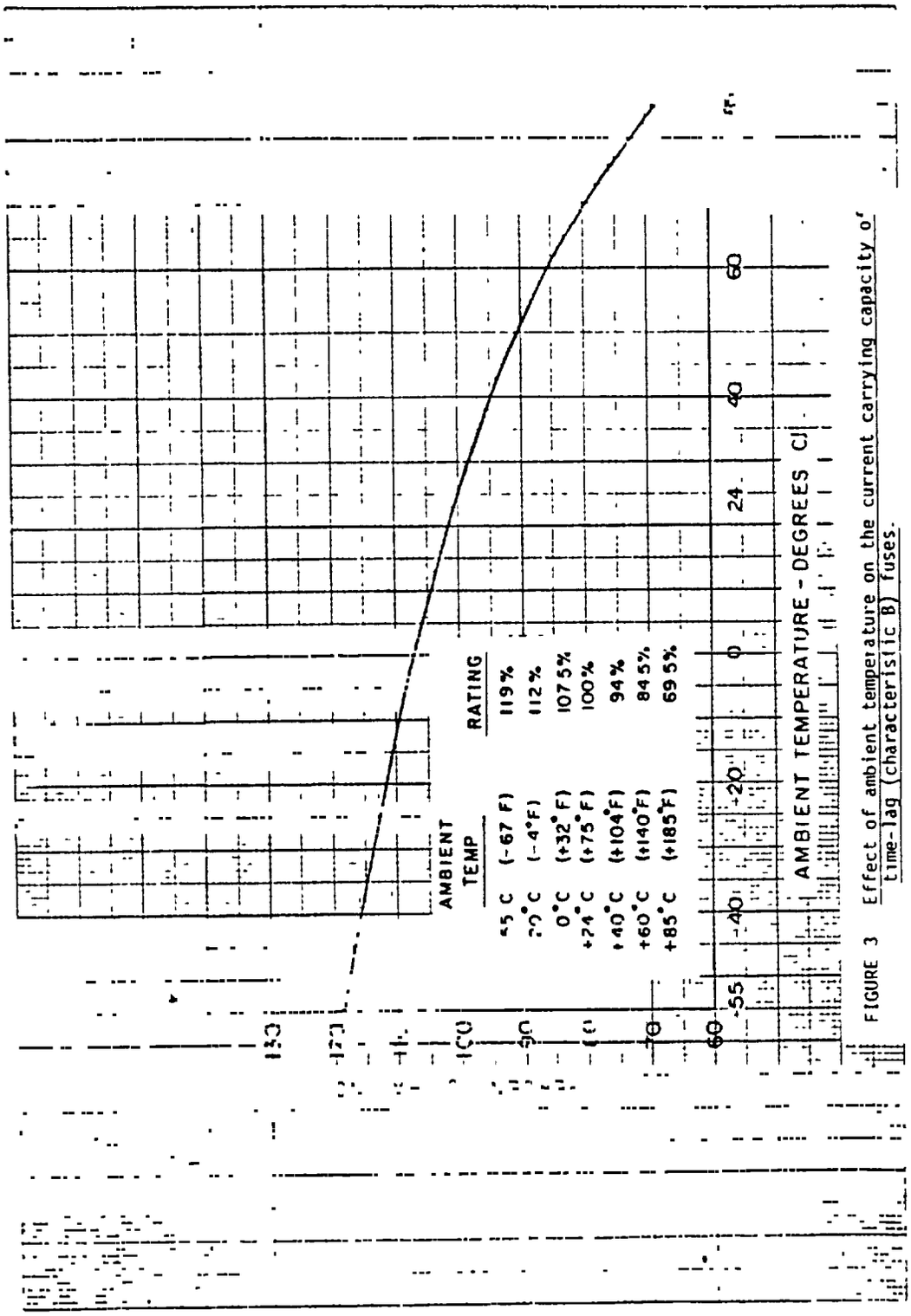


FIGURE 3 Effect of ambient temperature on the current carrying capacity of time-lag (characteristic B) fuses.

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Cross-reference: For applicable cross-reference, see table IV. The existing stocks of superseded items may be used. When exhausted, the superseding parts shall be used.

TABLE IV. Cross-reference.

Superseding number MIL-F-23419/9	Superseded number MIL-F-23419/6
FM09B250V1/100A	FM06-250V 1/100A
FM09B250V1/32A	FM06-250V 1/32A
FM09B250V1/16A	FM06-250V 1/16A
FM09B250V1/10A	FM06-250V 1/10A
FM09B250V1/8A	FM06-250V 1/8A
FM09B250V15/100A	FM06-250V 15/100A
FM09B250V3/16A	FM06-250V 3/16A
FM09B250V2/10A	FM06-250V 2/10A
FM09B250V1/4A	FM06-250V 1/4A
FM09B250V3/10A	FM06-250V 3/10A
FM09B250V3/8A	FM06-250V 3/8A
FM09B250V4/10A	FM06-250V 4/10A
FM09B250V1/2A	FM06-250V 1/2A
FM09B250V6/10A	FM06-250V 6/10A
FM09B250V7/10A	FM06-250V 7/10A
FM09B250V3/4A	FM06-250V 3/4A
FM09B250V1A	FM06-250V 1A
FM09B250V1-1/4A	FM06-250V 1-1/4A
FM09B250V1-1/2A	FM06-250V 1-1/2A
FM09B250V1-6/10A	FM06-250V 1-6/10A
FM09B250V2A	FM06-250V 2A
FM09B250V2-1/2A	FM06-125V 2-1/2A
FM09B250V2-8/10A	FM06-125V 2-8/10A
FM09B250V3A	FM06-125V 3A
FM09B250V3-2/10A	FM06-125V 3-2/10A
FM09B125V4A	FM06-125V 4A
FM09B125V5A	FM06-125V 5A
FM09B125V6-1/4A	FM06-125V 6-1/4A
FM09B125V7A	FM06-125V 7A
FM09B125V8A	FM06-125V 8A
FM09B125V10A	FM06-125V 10A
FM09B125V12A	FM06-125V 12A
FM09B125V15A	FM06-125V 15A
FM09B125V20A	FM06-125V 20A
FM09B125V25A	FM06-125V 25A
FM09B125V30A	FM06-125V 30A

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, SH
Air Force - 11, 99
DLA - ES
NS
NASA - NA

User activities:
Army - AR, ME
Navy - MC, OS
Air Force - 19

Preparing activity.
Navy - EC

Agent.
DLA - ES

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